

Python Programming - 2301CS404

Lab - 3

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01) WAP to check whether the given number is positive or negative.

```
In [4]: num = int(input("Enter a number:"))
        if(num>0) :
            print("num is positive.")
        elif(num<0) :
            print("num is negative.")
        else :
            print("num is zero.")
```

num is positive.

02) WAP to check whether the given number is odd or even.

```
In [6]: num = int(input("Enter a number:"))
        if(num%2==0) :
            print("num is even.")
        else :
            print("num is odd.")
```

num is even.

03) WAP to find out largest number from given two numbers using simple if and ternary operator.

```
In [8]: a = int(input("Enter first number:"))
        b = int(input("Enter second number:"))
        if(a>b) :
            print(f"{a} is largest.")
        else :
            print(f"{b} is largest.")
        ans = a if a>b else b
        print(f"{ans} is largest.")
```

8 is largest.
8 is largest.

04) WAP to find out largest number from given three numbers.

```
In [19]: a = int(input("Enter first number:"))
b = int(input("Enter second number:"))
c = int(input("Enter third number:"))
if(a>b and a>c) :
    print(f"{a} is largest.")
elif(b>a and b>c) :
    print(f"{b} is largest.")
else :
    print(f"{c} is largest.")
```

5768907 is largest.

05) WAP to check whether the given year is leap year or not.

[If a year can be divisible by 4 but not divisible by 100 then it is leap year but if it is divisible by 400 then it is leap year]

```
In [12]: year = int(input("Enter the year:"))
if(year%4==0 and year%100!=0 or year%400==0) :
    print(f"{year} is leap year.")
else :
    print(f"{year} is not leap year.")
```

2025 is not leap year.

06) WAP in python to display the name of the day according to the number given by the user.

```
In [14]: day = int(input("Enter the day:"))
match day :
    case 1 :
        print("Sunday")
    case 2 :
        print("Monday")
    case 3 :
        print("Tuesday")
    case 4 :
        print("Wednesday")
    case 5 :
        print("Thursday")
    case 6 :
        print("Friday")
    case 7 :
        print("Saturday")
    case _ :
        print("Invalid number")
```

Tuesday

07) WAP to implement simple calculator which performs (add,sub,mul,div) of two no. based on user input.

```
In [17]: a = int(input("Enter first number:"))
b = int(input("Enter second number:"))
operation = input("Enter the operation(+,-,*,/):")
if(operation=='+') :
    print(f"Sum of {a} and {b} is {a+b}.")
elif(operation=='-') :
    print(f"Subtraction of {a} and {b} is {a-b}.")
elif(operation=='*') :
    print(f"Multiplication of {a} and {b} is {a*b}.")
elif(operation=='/') :
    if(b!=0) :
        print(f"Division of {a} of {b} is {a/b}.")
    else :
        print("Division is not possible.")
else :
    print("Invalid number or operation.")
```

Multiplication of 15 and 4 is 60.

08) WAP to read marks of five subjects. Calculate percentage and print class accordingly.

Fail below 35

Pass Class between 35 to 45

Second Class

between 45 to 60

First Class between 60 to 70

Distinction if more than 70

```
In [18]: a = int(input("Enter the marks of sub-1:"))
b = int(input("Enter the marks of sub-2:"))
c = int(input("Enter the marks of sub-3:"))
d = int(input("Enter the marks of sub-4:"))
e = int(input("Enter the marks of sub-5:"))
percentage = (a+b+c+d+e)/5
if(percentage>70) :
    print("Distinction")
elif(percentage>60) :
    print("First Class")
elif(percentage>45) :
    print("Second Class")
elif(percentage>35) :
    print("Pass Class")
else :
    print("Fail")
```

Second Class

09) WAP to find the second largest number among three user input numbers.

```
In [24]: a = int(input("Enter first number:"))
b = int(input("Enter second number:"))
c = int(input("Enter third number:"))
if(a>=b and a<=c) or (a>=c and a<=b) :
    print(f"{a} is secondlargest.")
elif(b>=a and b<=c) or (b>=c and b<=a) :
    print(f"{b} is secondlargest.")
else :
    print(f"{c} is secondlargest.")
```

34 is secondlargest.

10) WAP to calculate electricity bill based on following criteria. Which takes the unit from the user.

- a. First 1 to 50 units – Rs. 2.60/unit
- b. Next 50 to 100 units – Rs. 3.25/unit
- c. Next 100 to 200 units – Rs. 5.26/unit
- d. above 200 units – Rs. 8.45/unit

```
In [25]: a = float(input("Enter the used units:"))
bill = 0
if(a>=1 and a<=50) :
    bill = a*2.60
if(a>50 and a<=100) :
    bill = 50*2.60 + (a-50)*3.25
if(a>100 and a<=200) :
    bill = 50*2.60 + 50*3.25 + (a-100)*5.26
if(a>200) :
    bill = 50*2.60 + 50*3.25 + 100*5.26 + (a-200)*8.45
print(f"{bill}")
```

214.5

In []: